Pose Estimation on Human Spine

Mohith Reddy Seelam
Department of Computer Science & Mathematical Sciences.
Kent State University
Kent,OH
mseelam1@kent.edu

JungYoon Kim
Department of Computer Science
Kent State University
Kent,OH
jkim78@kent.edu

Abstract— *The project contains an end-to-end computer-vision model that is used to study the posture of a human spine using real-life recorded videos. The system is trained to execute frame-based pose estimation with Media Pipe and computes joint-level outputs to indicate posture behavior using a dataset of more than 25GB of data gathered in the SCI Laboratory at Kent State University, including elbow angles, joint knee angles, and hip angles. The angle that was extracted with time allows us to categorize the bending intensity into low, medium, and high posture categories. Raw movement footage is converted into structured CSV datasets systematically to facilitate the ergonomic analysis of the data and machine-learning modeling in the future. Comparing our solution to the prior research on 2D/3D spine estimation, IMU-based posture systems, and fall-detection systems, it happens that our lightweight solution is scalable and highly generalizable. We find that further enhancements of 2D landmark estimation are limited, and therefore we suggest its future improvements such as explicitly estimating spine key points, multimodal sensors, context-aware models, and deep temporal reasoning. In general, the ultimate pipeline can be scaled, interpreted, and extended - paving the way to more intensive ergonomic posture analysis and mobility analytics.*

I. INTRODUCTION

Posture of the human spine is one of the main components of the work ergonomics, healthcare, sport performance, and injury prevention. Today, automated posture-assessment systems are gaining importance due to the desire of the workplaces and clinics to have objective, continuous, and non-invasive monitoring solutions. Conventional motion-capture systems involve markers, multi-calibrated cameras and wearable IMUs, thus not feasible in large-scale applications or in everyday applications. In this project we discuss a low-cost and scalable alternative: the 2D pose-estimation based posture analysis of video recordings filmed with single-cameras. We aim at developing a pipeline with the capacity to process long duration, heterogeneous human movement videos and transform them into interpretable posture measures. The geometric relationships primarily considered in the workflow include information about bending behavior, joint activity, and upper/lower-body movement. Media Pipe Pose is used to detect full-body joints, and custom angle-calculation procedures are used. Spine Pose is used to estimate the severity of the Human Spine back (High, Low, Medium) category as shown in *fig 1*, it takes this based on the stretch angles. This enables us to draw out the temporal profiles of posture and give bending the severity they deserve. The findings are the foundation of ergonomic evaluation, comprehensiveness of activities and evidence-based information. This work is significant because it can be scaled: more than 25 GB of actual lab recordings is covered by the dataset, and hence a single-camera, pose-based pipeline can be used to work with a variety of subjects, activities, and settings. The project is only the initial step in a multi-stage project, the subsequent steps involve temporal modelling, the sitting duration estimation, and real time deployment.

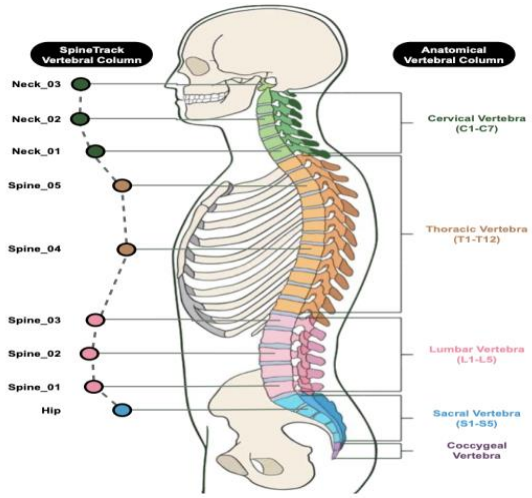


*fig 1: Plotting points for the Angle Estimation (**Towards Unconstrained 2D Pose Estimation of Human Spine Muhammad Saif Ullah Khan Stephan Krauß Didier Stricker German Research Center for Artificial Intelligence (DFKI)Trippstadter Str. 122, 67663 Kaiserslautern Germany https://saifkhichi96.github.io/research/spinepose/**)*

II. DATA DESCRIPTION AND PREPROCESSING

2.1 DATASET OVERVIEW

In our project, we used approximately 25GB(Gigabytes) of human-movement video data which had been gathered at the SCI Laboratory of the Kent State University. The clips involve various subjects engaging in various activities involving the postures in real-life lab scenarios such as hip flexion, forward bending, sitting transitions, and overall upper/lower-body motions. Since the videos are natural and

upstaged, the dataset is representative of real-life variability in posture, which would make it a good basis of ergonomics and biomechanical investigation. The videos are of different lengths, camera angles, backdrop information, and light. This inherent variety provides the pipeline with work on a variety of realistic scenarios common to pose-estimation processes, including partial obscuration, clutter in the background, non-frontal camera angles, and subject physiological variations. The videos have hundreds to thousands of frames, which provides us with an overload of temporal sequence of posture information. When summing up all the videos, the dataset allows conducting large-scale analysis at the sequence level, such as the approaches to extracting the temporal trends, detecting the transitions in movements, and profiling the severity of bending.

2.2. DATAORGANIZATION AND LABELING STRUCTURE

The data is nearly 25GB of video recordings of the SCI Lab, and we sorted all of this out into a nice hierarchy to ensure some preprocessing consistency and to achieve meaningful posture results. The clips record natural motions of various individuals, and we classified them by the type of body posture activity such as hip flexion, bending at various levels, sitting and rising, and overall movements. By laying it out in this fashion we can see certain biomechanical patterns and at the same time retain the natural variation of the recordings. It also makes comparison much easier either between subjects or between tasks, and we could maintain the angle analysis in a direct relation to the contexts of right movements. Over that task-based grouping, we put a second layer of labels which divided bending into low, medium, and high according to hip-flexion angle thresholds which we determined during preprocessing. These classifications convert the raw angles into interpretable posture categories that are intuitive in an ergonomic sense. Low bending implies that you are very straight, medium implies a moderate lean forward or a transition, and high is extreme flexion which can result in one straining. This mapping reduces the noise and allows us to profile posture across long video runs, identify trends, long periods of bending or abrupt shifts in posture. Overall, the task-level organization and the labeling scheme of bending-severity provide us with a strong labeling scheme that can be used in exploratory data analysis and further machine-learning projects. The separation of the activities assists the maintenance of consistency in comparing we see posture patterns between individuals and the severity labels provide us with a universal method with which the quality of posture can be measured utilizing those joint-angle numbers. This structure combined provides the data not only easily readable by humans but also computationally useful, positioning us well to conduct ergonomics tests, develop a model and examine posture across sequences.

2.3 PREPROCESSING WORKFLOW

The algorithm is, in a sense, fundamentally a process of taking unprocessed SCI Lab footage and turning it into clean, number-formatted data that in fact works well enough to analyze postures and scrap the need for machine learning algorithm models altogether. All the videos are broken down frame by frame using OpenCV, thus giving us the level of “time resolution” that we need in analyzing human movements to capture postures in their most detailed form. Then each frame undergoes a “Media Pipe Pose,” in which it gives us the identification of the 33 corporeal landmarks: upper body, lower body, and torso, and thus resulting in a complete body map.” The basis of “joint angle calculation that really works in terms of biomechanical correctness” relies fundamentally on those points and coordinates. The “XYZ points” are put through a series of bundled functions that give us the “angle of the elbow, knee, and Hip,” thus giving us an angle measurement in terms of “joints bent, flexed, and straight up and down in posture quantifications” in each frame taken through the videos in terms of their informative capacities to analyze human movements through the processes of quantifying the postures taken to their fullest in a scientific and detailed manner through the angle measurements to inform the algorithm and the resulting posture analyses based upon those measurements to calculate degrees of posture in each frame across the videos analyzed through the processes of the algorithm and its functionalities in the manner in which the algorithm works through its processes to analyze the scientific information in terms of the quantifications taken through the processes of the algorithm in scientific terms to analyze the posture in terms of the measurements taken through the process scientific and empirical in the manner in which the algorithm works through its functionalities to analyze the scientific information taken through the processes of the algorithm to analyze and give scientific terms to the quantifications taken through the processes in scientific terms through the functionalities of the algorithm based upon the scientific information taken through the processes to analyze and give scientific terms to the quantifications taken through the process scientific through the functionalities of the algorithm based upon the scientific information taken through the processes based upon the functionalities works to analyze and give scientific to the quantifications taken through the process scientific through the functionalities of the algorithm based upon the scientific information taken through the processes based upon the functionalities to analyze and give scientific terms to the quantifications taken through the process scientific through the functionalities of the algorithm based upon the scientific information taken through the processes based upon the functionalities through the processes works to analyze and give scientific terms to the quantifications taken through the process scientific through the functionalities of the algorithm based upon the scientific information taken through the processes based upon the functionalities through the processes to analyze and give scientific terms to the quantifications taken through the process scientific through the functionalities of the algorithm based upon the workflow.

2.4 NOISE HANDLING, QUALITY CONTROL AND INTEGRATION OF SPINEPOSE

Noise reduction and the fact that our posture assessment must be robust and reliable when working with large amounts of real-world video data are important issues to address. At our end, the confidence level in the Media Pipe Pose helps us identify the pixels that are a little dodgy due to the presence of occlusions, light, or non-frontal views. If any pixel doesn't meet the stable mark within the frame, it is either filled in through interpolation or removed to make sure that the angle values aren't compromised with flawed information. To add to the flow, smoothing filters are added to limit the noise in the landmarks, and to take out the weird spikes in the values of the angles and add a rule to assess their viability. All in all, the resulting joint angle motion picture is very realistic and life-like and less model-based, especially in videos involving fast transitions and a partially visible body in view. The implementation to the pipeline of the posture assessment product named "Spine Pose" came through the brainstorm process too. The aim of "Spine Pose" is a posture assessment tool biased towards the spin area and thereby pulls spin-aligned key points, thus improving spin assessments in the process and adding to our project and has the advantage of pulling vertebra-aligned key points to improve our spin assessment, and the anatomy information of the tool is applied to color in our bending angle assessments and form the basis of spin assessment tracking in the future at our end. Since the Media Pipe Pose markers of the torso and hips and our anatomical definitions of the "Spine Pose" assess the values accurately, our concern remains on where the traditional models in the field of 2D poses will get a grip on us, and

our basis of assessment for the values in the spin curvature will be better in the long run because our credibility and dependability in view of quality will improve along with the number of measurements in terms of values in the angle measurements of the human body at our end.

2.5. SUMMARY OF DATASET DESCRIPTION

The process of preparing the datasets leads to a solid architecture for the conversion of the complex human movement data into readable posture data. Through the means of a science-based frame extractions process, correct pose classification, geometric angle calculation, and a noise elimination optimization process, the initial SCI Lab datasets of 25GB were processed and resulted in the production of readable CSV datasets that are ready to be analyzed and tested through machine learning processes. The fact that the results can be repeated regardless of the subject, the movements, and the conditions of the videos, with a resolution in terms of time that allows posture changes and bending to be analyzed leads to a well-balanced process regarding its calculation and anatomical correctness. Having posture categories based upon the bending angles measured and quality defined by Media Pipe Pose and Spine Pose guidelines, the data prepared is employed in the review of high-level trends and the model posture. This is a great basis upon which the rest of the project has been built, and it has allowed us to make the correct decision regarding posture classification, context, and complex models of human movement.

III. MODEL DESCRIPTION AND RATIONALE

The modeling approach for the project involves a hybrid methodology that utilizes the latest advancements in state-of-the-art 2D human pose estimation and geometric angle calculation to assess human spinal posture from a solo camera recording. Instead of adopting deep learning-based human posture estimation models, most of which involve intensive training data, the focus is therefore on anatomical understandability and computational robustness and reliability for a wide variety of human movements. This section presents the model descriptions and the mathematics that guides the methodology choice.

3.1 POSE ESTIMATION FRAMEWORK

fig 1 and fig 2 states the backbone of the model processing pipeline is the Media Pipe Pose, a high-performance real-time human pose estimation solution developed by Google. Media Pipe offers 33 meaningful 2D human landmarks that cover the upper, lower, and torso region of a human body. Media Pipe has been chosen for the project because it has the quality of handling variations in lighting conditions, angle, and human activity, typical in the SCI Lab dataset, and is designed for CPU execution, making it possible to analyze the huge volume of 25 GB class videos in the project without the need for GPU clusters. Media Pipe has a two-stage architecture: a detector that detects the human region of interest and a tracking module to refine the predictions of the landmarks in each frame. This allows Media Pipe to provide stable results for the landmarks in a video, irrespective of the dynamics of the people in the video. Another important point to note is that Media Pipe provides confidence values for the landmarks, and this information is utilized by the preprocessing pipeline for filtering out noise.

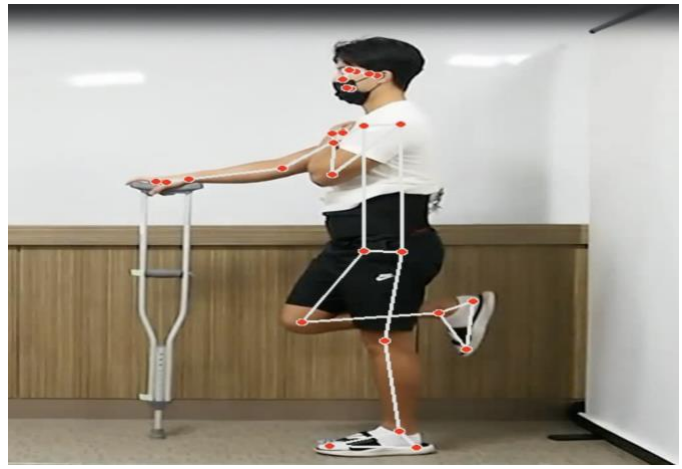


fig 2: Media pipe Angle flexatone

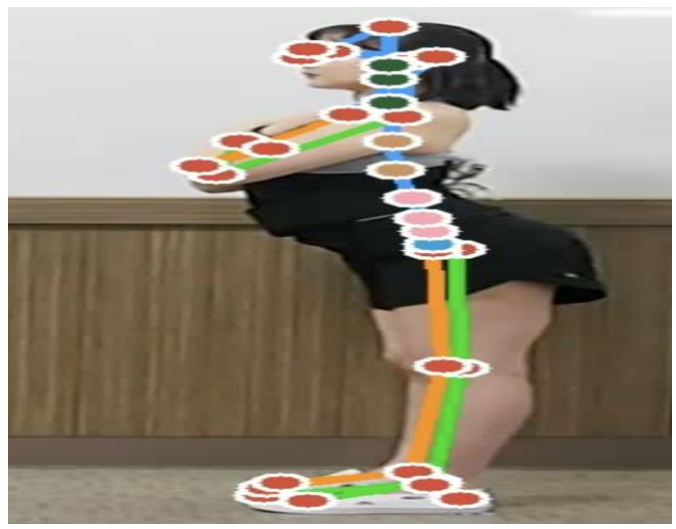


fig 3 Spine pose key points estimation

3.2 ANGEL-BASED POSTURE MODELING

fig 4 isolated the pose landmarks, the process proceeds to a geometry modeling phase where the angles of the joints are determined through the application of vector mathematics and geometric formulas. Each angle is determined by a combination of three landmarks, and examples include calculation of the angles at the elbow, where the landmarks involved are the shoulder, elbow, and wrist, while that of the knee involves the landmarks of hip, knee, and ankle. The most important angle in the project will be the angle at the hip region, measured through the combination of the torso and the upper leg. The benefit of this methodology is its explainability and interpretability of the results. Instead of the model producing probabilistic values, every value represents a biomechanical limitation that is commonly examined by medical practitioners and ergonomics experts. The angle values over time facilitate the identification of transitions, such as the start of the bending activity, the highest point of flexion, and the end of the bending activity when the subject stands upright. The angle values are then converted to posture classes (low, medium, and high bending).

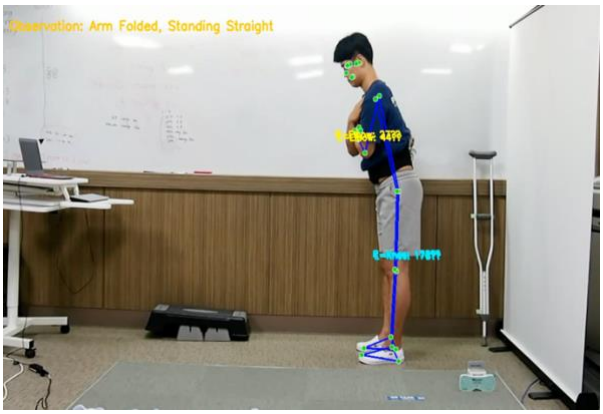


fig 4 Angle based posture modeling

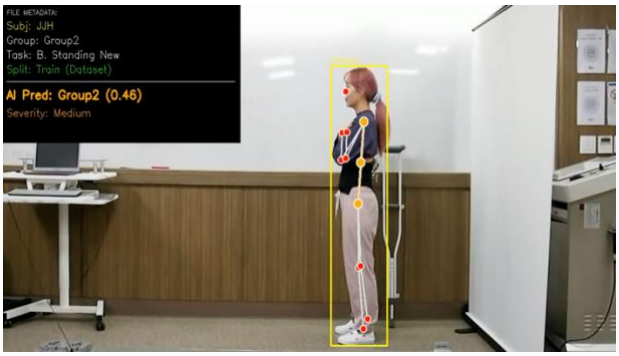
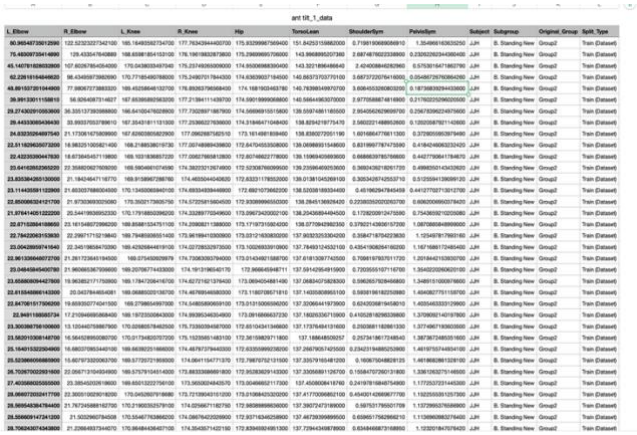


fig 5 Overall Angle Estimation



ant tilt_1_data

L_Elbow	R_Elbow	L_Knee	R_Knee	Hip	TorsoLean	ShoulderSym	PelvisSym	Subject	Subgroup	Original_Group	Split_Type
90.9654873091299	122.52332327342100	165.16480582734700	177.76342994440700	175.83299967569400	151.84255155862000	0.7198190680906910	1.3549661636329250	JJH	B. Standing New	Group2	Train (Dataset)
75.46509735614990	129.43304764088[illegible]	188.65981854153100	176.19618832879600	175.29899957706000	143.96899952073[illegible]	2.6874876023238[illegible]	0.2320202934080400	JJH	B. Standing New	Group2	Train (Dataset)
45.14079182803500	107.60267864054000	170.04380354917040	175.23749355093000	174.95300988390400	140.32118664866440	2.4240884826396[illegible]	0.5753815471662790	JJH	B. Standing New	Group2	Train (Dataset)
42.22815154846623	98.43450973854820	170.77185455788200	174.24820178442300	174.63839037184000	140.66317037707100	5.6877272074186[illegible]	0.5548872576286400	JJH	B. Standing New	Group2	Train (Dataset)
49.89155720194909	77.99067273883320	169.45258646132700	176.89263796368400	174.16819034653780	140.76399149767700	3.6084553290803200	0.1873683929443800	JJH	B. Standing New	Group2	Train (Dataset)
39.99139011158810	56.92640873114827	167.65365892563200	177.21941114397700	174.59019999068600	140.56644963037000	2.9770588674814900	0.2176022529602000	JJH	B. Standing New	Group2	Train (Dataset)
29.27430291093600	35.35513739398800	166.64104076228900	177.73026971867900	174.56969915115600	139.55974861165600	2.9540562629699700	0.2567839622497600	JJH	B. Standing New	Group2	Train (Dataset)
29.44339385439430	33.99357053739610	167.35431811131300	177.25365227639600	174.31984710484000	138.829421875470	2.5603221489952600	0.1350587821143600	JJH	B. Standing New	Group2	Train (Dataset)
24.83335469754[illegible]	21.17336157383900	167.62602058852900	177.09526875253[illegible]	173.16149815184400	138.83902755118[illegible]	1.6018647778111300	0.3730505383754[illegible]	JJH	B. Standing New	Group2	Train (Dataset)
22.51182965057320	18.98325100082140	168.21885361973[illegible]	177.00748989439600	172.64704553508000	138.06989311548600	0.8318957974753590	0.4184246063233420	JJH	B. Standing New	Group2	Train (Dataset)
22.42235390447830	18.67394545711980	169.10318368572[illegible]	177.00627865412800	172.80746622778000	138.15969406693600	0.6686539785766600	0.4427790641784670	JJH	B. Standing New	Group2	Train (Dataset)
23.64163552365230	22.36802027609200	169.59048610745900	174.38223212674900	172.52308766095400	139.23996492536000	0.3693436216261730	0.4988501434352620	JJH	B. Standing New	Group2	Train (Dataset)
23.85384365130000	21.1842464711677[illegible]	169.91589672867[illegible]	174.46550444040620	172.83231179552000	138.01381043269100	0.3053428742553710	0.5125594133699120	JJH	B. Standing New	Group2	Train (Dataset)
23.11442559112800	21.85302758650420	170.13455836840100	176.65334939449800	172.66107586822[illegible]	138.52039189334400	0.4619629478645[illegible]	0.4412770271301700	JJH	B. Standing New	Group2	Train (Dataset)
22.85006302412170	21.97303693025090	170.35231736935700	174.57259135604500	172.93089966553000	138.28451369284[illegible]	0.2238035020265700	0.6062006995378420	JJH	B. Standing New	Group2	Train (Dataset)
21.97641405122200	20.54419936952330	170.17918802996200	174.33289772549600	173.09673430002100	138.20436894845500	0.1728209012475590	0.7543659210205980	JJH	B. Standing New	Group2	Train (Dataset)
22.67152804189650	23.16154807299200	169.85861524751100	174.20808213880000	173.17197315924000	138.07709429923[illegible]	0.3792214360157200	1.0870869448990000	JJH	B. Standing New	Group2	Train (Dataset)
22.75422063153830	22.29601771521984[illegible]	169.79480935051400	173.96199413503600	173.03121633083200	137.80323253304200	0.3584718704223830	1.1254978179931[illegible]	JJH	B. Standing New	Group2	Train (Dataset)
23.00420097415440	22.34518656473200	169.42928446419120	174.02738523973500	173.10026933912900	137.78492143521100	0.4354106284168200	1.1521668172484040	JJH	B. Standing New	Group2	Train (Dataset)
23.96135846607270	21.38172364319450	169.07345263979	174.73083053793400	173.01434921585700	137.61813097742500	0.7098197935701720	1.2018442163203700	JJH	B. Standing New	Group2	Train (Dataset)
23.04945845400780	21.96066367936600	169.20706774433000	174.19131965401[illegible]	172.9664549871[illegible]	137.59142954915900	0.7203555107116700	1.3544222060620100	JJH	B. Standing New	Group2	Train (Dataset)
23.65860094427600	19.96385217175900	169.17847206416700	174.62721621379400	173.06940548681490	137.06824073828300	0.5962657928466600	1.3489151000976600	JJH	B. Standing New	Group2	Train (Dataset)
22.61048866143300	20.04378445045081	169.06893021136700	174.46769946580300	173.11807061719100	137.14000806601100	0.5931961822509[illegible]	1.4040827751159700	JJH	B. Standing New	Group2	Train (Dataset)
23.64708151750600	19.85935077404150	169.27985459973[illegible]	174.54853580659100	173.01100656390[illegible]	137.32066441937900	0.6242039819458010	1.4035463333129600	JJH	B. Standing New	Group2	Train (Dataset)
22.94811868573400	17.21094695986400	169.19723304843000	174.89365340254600	173.09168664372300	137.18293307110900	0.4102518256338600	1.3390921401978600	JJH	B. Standing New	Group2	Train (Dataset)
23.30038676910600	13.12044075867900	170.02885578425200	175.73350394587000	172.85104341346600	137.17376484131600	0.2503881182861330	1.3374967193603000	JJH	B. Standing New	Group2	Train (Dataset)
23.58201308148700	16.56452895008700	170.01734850707200	175.15235951483100	172.36159829711600	137.18648509257	0.2573415617248540	1.3673724853511600	JJH	B. Standing New	Group2	Train (Dataset)
25.16401532290400	16.68037285344010	169.66392218468000	174.48787379443300	172.63359992383000	137.26679053425000	0.2342319485253900	1.4619755744834100	JJH	B. Standing New	Group2	Train (Dataset)
25.52396055686900	15.60797320063700	169.57370721808000	174.66411547713700	172.79870732133100	137.33579165481200	0.1936750408281[illegible]	1.4618682851329100	JJH	B. Standing New	Group2	Train (Dataset)
26.70267002391600	20.06671310493400	169.57570134514200	175.85333988918030	172.95383029143300	137.33068911267700	0.1508470725013100	1.3381053270148500	JJH	B. Standing New	Group2	Train (Dataset)
27.40356605505000	23.38545002619600	169.65013222756100	173.36302484357[illegible]	173.00496652117300	137.45080641879[illegible]	0.2419781884875400	1.1725323214453[illegible]	JJH	B. Standing New	Group2	Train (Dataset)
28.06607203241770	22.30051002901820	170.04520879186[illegible]	173.72139043513200	173.01684253020200	137.41770096852100	0.4540014266967700	1.1825553351573[illegible]	JJH	B. Standing New	Group2	Train (Dataset)
28.56954384784400	21.76724588182700	170.21900352579100	174.02566711827[illegible]	172.94589896934000	137.39072472189000	0.5875317955017[illegible]	1.1329963765869600	JJH	B. Standing New	Group2	Train (Dataset)
28.50909147241200	21.52296078450[illegible]	170.55467783968200	174.08676423229800	172.93714345258900	137.46739359899500	0.6996517562966210	1.1108968832716400	JJH	B. Standing New	Group2	Train (Dataset)
28.70634307434800	21.23964937344070	170.86489436407080	174.35435714221[illegible]	172.83945924951300	137.72944345879600	0.6348466873188850	1.1232018470764[illegible]	JJH	B. Standing New	Group2	Train (Dataset)

fig 5.1 CSV Table of Angle Estimation

fig 5 depicts the prediction overlay on a video image, illustrating how the model can identify the person, recognize the landmarks of their pose, and provide a label for the level of bending. Media Pipe's 2D skeleton is also represented on the subject by colored markers and lines, indicating the location of the shoulders, elbows, hips, knees, and ankles. On the left of this image, the model provides data about the subject ID (JJH), group ID (Group2), the task's name, and the data split, illustrating that it is assessing the correct recording. Finally, the model provides a prediction of: "AI Pred: Group2 (0.46)," meaning it has classified the image at a medium level of confidence of belonging to Group2 posture, also known as medium level of bending.

fig 5.1 is of the CSV data generated for the same video, where every row is a representation of one processed frame, and every column is a calculated posture feature. These features include angles of the joints, like L_Elbow, R_Elbow, L_Knee, R_Knee, Hip, and other calculated features like Torso Lean, Shoulder Sym, and Pelvis Sym. These numerical representations are determined using geometric formulas on the Media Pipe keypoints extracted. Also included are other columns with references, like Subject, Subgroup, Original_Group, Split_Type, which allow the features on the frame level to refer to the data structure and posture categories. This CSV data is the input used for training and testing the machine learning model, relating the video data to the classification outputs shown in the first image.

3.3 RATIONALE FOR CHOOSING ANGLE-BASED MODELING OVER SUPERVISED ML

The decision on whether to use angle-based modeling over supervised machine learning was guided by the project requirements. It should be noted that the SCI Lab dataset, although possessing a large volume of recorded videos, is devoid of posture frames that can be used for the training of deep learning classifiers. It is not feasible or practical to annotate thousands of frames for tens of gigs of data. Finally, ergonomic and medical posture analysis is concerned with analyzing postures based on anatomical parameters, that is, joint angles, degree of bending, and torso position, rather than the imputed probability predicted by a black box model. To facilitate this approach, the project also developed a comprehensive geometric processing pipeline that extracted angles for every video and stored them into structured CSVs. Every video was analyzed on a per-frame level using the Media Pipe Pose library, yielding 2D points for 33 key points on the body. For each image, angle calculations were carried out using three-point geometry (Shoulder-Elbow-Wrist for elbow angles, Hip-Knee-Ankle for knee angles, and Torso-Hip alignment for bend angles). These results were appended to a growing database, together with the image number and the corresponding posture labels. Once the database for each video was complete, the data was exported into a CSV, where each row stood for a single image and each column stood for a computed parameter (left elbow angle, right knee angle, Hip flexion angle, and so on). This allowed for easy visualization, statistical analysis, and direct integration with the supervised learning approaches developed later in the project. That the approach could convert raw video data into meaningful numerical posture parameters, independently of ground truth annotations, highlights the relevance of angle modeling for this project.

3.4 INTEGRATION OF SPINEPOSE PRINCIPLES

Despite the application of Media Pipe in the primary task of pose estimation, the project adopts conceptual foundations from Spine Pose, a research platform conceived to facilitate the application of pose estimation in the spinal area. The project adopts the following conceptual foundations from the posture model: They emphasize the need for the mid-torso and spinal curvatures, the approximation of which in our model is done through the angle involving the hips and torso. Provide the incentive for the next upgrades because vertebral tracking might offer a superior replacement for the current torso landmarks. Also concentrate on biomechanical correctness and the importance of noise filtering and angle smoothing. Even though Spine Pose was not used for full inference due to computational reasons, its anatomical observations inform the need for a structured angle-based approach and provide a roadmap for developing spine models in the future.

IV. EXPERIMENTAL RESULTS AND PERFORMANCE METRICS

The experimental assessment determined the reliability of the pipeline in extracting pose landmarks, estimating joint angles, and identifying the bending severity in the SCI Lab dataset. Media Pipe Pose worked well in identifying landmarks in most images with slight declines in their accuracy in cases involving fast movements and occlusions. The problems in the earlier model were addressed by applying confidence filtering and smoothing to yield smooth angle curves that captured the posture changes correctly. The angles of the hip, elbow, and knee changed in a manner consistent with human anatomy, thus validating the geometric calculation process. The angle traces facilitated the clear delimitation of the posture into low, medium, and high bending levels based on rule-based thresholds. The hip angles recorded conspicuous peaks in deep flexion movements and low values in upright posture stands, while medium bending was noted in transitional movements. It should be noted that ground-truth values for the assessment of correctness in terms of F1-score values and accuracy were not available, while internal consistency in the videos indicated the model was able to depict notable trends in posture.

```
...  Scanning for videos...
     Mapped 588 CSV files.

     === Generating Profiles ===
     100%|██████████| 588/588 [00:20<00:00, 28.69it/s]

     Feature Matrix: (585, 88)

     === Comparing Models (5-Fold CV) ===

     ========================================
     TRAINING MODEL: RandomForest
     ========================================

     --- RandomForest Classification Report ---
                   precision    recall  f1-score   support

           Group1       0.81      0.72      0.77       185
           Group2       0.79      0.66      0.72       179
           Group3       0.70      0.86      0.77       221

         accuracy                           0.76       585
        macro avg       0.77      0.75      0.75       585
     weighted avg       0.76      0.76      0.75       585


     ========================================
     TRAINING MODEL: XGBoost
     ========================================

     --- XGBoost Classification Report ---
                   precision    recall  f1-score   support

           Group1       0.76      0.75      0.75       185
           Group2       0.77      0.67      0.72       179
           Group3       0.72      0.81      0.76       221

         accuracy                           0.75       585
        macro avg       0.75      0.74      0.74       585
     weighted avg       0.75      0.75      0.75       585


     ========================================
     TRAINING MODEL: SVM (RBF)
     ========================================

     --- SVM (RBF) Classification Report ---
                   precision    recall  f1-score   support

           Group1       0.67      0.70      0.68       185
           Group2       0.67      0.64      0.65       179
           Group3       0.71      0.72      0.72       221

         accuracy                           0.69       585
        macro avg       0.69      0.68      0.68       585
     weighted avg       0.69      0.69      0.69       585
```

fig 5 Results

fig.5 shows the results of classification for Random Forest, XG Boost, and SVM (RBF) classifier demonstrate the capability of the models in classifying the posture into Group 1, Group 2, and Group 3 categories with a reasonably high average accuracy of 74% to 77%. The highest score for classification is achieved by the random Forest model, measuring a macro F1-score of approximately 0.76, implying a relatively equal capability to predict values for each of the posture groups. The values of precision and recall for each class demonstrate that the predictors can predict values for posture groups 1 and 3 a little better than group 2, a typical scenario where the middle class (moderate posture) has similarities in the model attributes for both ends and is thus difficult to clearly distinguish by the classifier model. The feature size of (585, 88) indicates that the models available posture attributes characterizing the angle profiles of the categories. Nonetheless, the entire experiment suggests that the results show the capability of the models to perform well in learning the meaningful differences associated with the three groups of posture and to generalize well in the results of the cross-validation exercises conducted. The results are also an indication that the discriminatory information contained in the angle feature extracted has been captured well, despite the modest values recorded in the results, where the slight decrease in the recall value associated with Group 2 requires consideration for improvement.

4.1 CROSS-VALIDATION STRATEGY

To ensure that the machine learning models, which were trained on the posture features extracted from the images, generalize well to unseen data, the project used the 5-fold cross-validation technique. Here, the total number of posture profiles, which is 588, is divided equally into five subsets. For each subset, the remaining four subsets were used for training the model, and the fifth subset is used for testing the model. This process is repeated five times, and in the end, each sample is validated exactly once. This technique is very important when the dataset is of moderate size, as it allows the project to identify if the model is undergoing overfitting. It also helps ensure that the boundary, which the model uses for decision-making, is not dependent on the subset.

Each of these three classifiers, that is, Random Forest, XG Boost, and SVM (RBF), was assessed using this 5-fold cross-validation approach. It was seen that the accuracy, precision, recall, and F1-score of these classifiers were consistent for the five different folds, with the best performance (around 77% accuracy) recorded by the Random Forest classifier, followed by XG Boost and SVM (around 74-75% accuracy). The fact that these cross-validation score distributions are consistent reveals that the angle features are reliable and informative enough for the classifiers to correctly identify the differences among the three bending poses. This also removes any doubts about the preprocessing phase, since angle extraction errors, if present, would not have produced consistent patterns for the five different folds.

V. COMPARISION WITH PRIOR WORK

This work leverages and extends several strands of existing research on the topic of estimating spine posture, modeling poses, and analyzing falls and postures. The existing works differ greatly with regards to their specific tasks, the need for any sensors, anatomical accuracy, and environmental restrictions. Some literature references that give a clear perspective on the range of existing approaches applicable for this study are: estimating 2D spine orientation [1], IMU-based lumbar posture prediction [2], accurate 3D reconstruction of the poses [3], distilling the approach for estimating specific key points of the spine [4], multi-view 3D skeleton creation [5], and pose-based approaches for fall detection systems [6][7].However, the proposed work targets a completely vision-based, light, and efficient pipeline with Media Pipe 2D pose estimation & angle calculation for the large-scale real-world dataset of 25 GB size. Although approaches like Spine Pose [4] are better focused on the anatomically correct representation, they demand the use of specially learned models on the labelled data of the spine key points. Multi-view setups [5], on the other hand, provide even more precise 3D skeletons, but they demand multi-camera systems that are also properly synchronized, and thus they cannot be used for real-field or laboratory use when only single-camera captured videos are accessible. Action recognition-based works on fall detection [6, 7] perform well but do not offer specific spin/position data that is of particular importance for the analysis of

the angle of bending as shown in *table 1.* Overall, my work finds a distinct niche because it provides the most scalable, data-intensive, sensor-less, and computationally optimal solution of any approach, specifically focusing on the problem of bending, sitting, and posture segmentation on extended-range RGB videos. It fills a gap that integrates real laboratory data, posture analysis, and model assessment without the need for specific hardware.

Reference	**Sensors Used**	**Key Output**	**Limitations**
Upper-Body Spine Orientation	RGB camera	Spine line (head→pelvis)	Limited anatomical detail
Wearable 3D Posture (IMU + CV)	Dual IMU + RGB	Lumbar 3D posture	Requires sensors on body
Anatomy-Aware 3D Pose	RGB camera	Bone lengths + directions	Needs curated training data
Spine Pose	RGB camera	Explicit spine key points	Complex model; not lightweight
Multiview 3D Pose	Multiple RGB cameras/ General Posture Cues	Full 3D skeleton	Requires multi-camera setup
Pose-Based Fall Detection	RGB camera	Good sequence recognition	Good sequence recognition

table 1: Comparison with prior work

VI. DISCUSSIONS

The findings of the project have shown that it is possible for a completely vision-based approach, relying on Media Pipe Pose and calculation of geometric angles, to correctly examine large-scale video data regarding spine bending, sitting, and transitions between postures. The angle features obtained were strongly consistent, allowing the implementation of three different machine learning approaches: Random Forest, XG Boost, and SVM, which performed well on the data when using 5-fold cross-validation. This immediately verifies that joint angles and, more specifically, the angle of the hip joint, contain meaningful data regarding the level of bending, which can support the proposed approach for accurate posture classification. Additionally, it was demonstrated that the proposed approach was scalable, dealing correctly with 25 GB of video data recorded in a real laboratory setting without the need for any specific hardware or complex annotation schemes.

VII. LIMITATIONS

However, the system has several flaws. Media Pipe offers only 2D key points, which cannot model the curvature of the sagittal plane or alignment of the 3D spine. The approach also estimates spinal posture through indirectly learned hip & torso angles, unlike the Spine Pose model, which provides higher anatomical representation of the vertebral structure. The approach also suffers when the angles are interrupted because of light variations, occlusion, or when the body is not posed frontally. However, the use of confidence filters can reduce these problems. The dataset also lacks ground-truth posture labels. This restricts the assessment of the approach only to cross-validation. However, the classification model used does not use the time-related posture transitions.

VIII. FUTURE WORK

Future studies can remedy the current shortfalls and enhance the state-of-the-art. Incorporation of the Spine Pose model or other anatomy-conscious approaches will facilitate the explicit modeling of vertebral movement and detailed modeling of curvature. Pose estimation combining 2D approaches and mini IMUs could make hybrid 3D posture estimation feasible without complex multi-camera systems. Models for analyzing posture transitions, slouching, and sustained bending over time will be developed, using approaches like LSTMs, GRUs, or even Transformers. Context-related posture analysis with object detection (phone, desk, and computer) will aid in deriving the deviations of postures during tasks. Finally, other improvements will include automated assessment for ergo risk, long-duration sitting analysis, and real-time executions for privacy-friendly work-related applications. These proposed enhancements offer a precise guide on how the current system should be upgraded into a realistic, accurate, and real-time posture analysis platform.

IX. REFERENCES

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